

VENKATESH MISHRA

venkatesh.mishra@asu.edu | [in](#) Venkatesh Mishra | [G](#) VenkyMishra

RESEARCH INTEREST

Large Language Models (LLMs), Language Agents Reasoning and Evaluation, Post-training

EDUCATION

Arizona State University <i>Doctorate in Computer Science</i>	Tempe, Arizona August 2024 - Present
Arizona State University <i>Master of Science in Computer Science</i>	Tempe, Arizona Aug 2022 - May 2024
National Institute of Technology, Silchar <i>Bachelor of Technology in Electronics and Instrumentation</i>	Silchar, Assam, India Jul 2016 - Jun 2020

PUBLICATIONS

- Amir Saeidi, **Venkatesh Mishra** (co-first author), Souradeep Mukhopadhyay, Gaowen Liu, Ali Payani, Jayanth Srinivasa, Chitta Baral [FAMA: Failure-Aware Meta-Agentive Framework for Open-Source LLMs in Interactive Tool Use Environments](#). *Findings of Association for Computational Linguistics (ACL 2026)*
- Venkatesh Mishra**, Amir Saeidi, Satyam Raj, Mutsumi Nakamura, Jayanth Srinivasa, Gaowen Liu, Ali Payani, Chitta Baral [How Can Input Reformulation Improve Tool Usage Accuracy in a Complex Dynamic Environment? A Study on \$\tau\$ -bench](#). *Findings of Association for Computational Linguistics (EMNLP 2025)*
- Venkatesh Mishra**, Bimsara Pathiraja, Mihir Parmar, Sat Chidananda, Jayanth Srinivasa, Gaowen Liu, Ali Payani, Chitta Baral. [Investigating the Shortcomings of LLMs in Step-by-Step Legal Reasoning](#). *Findings of Association for Computational Linguistics (NAACL 2025)*
- Shrinidhi Kumbhar, **Venkatesh Mishra**, Kevin Coutinho, Divij Handa, Ashif Iquebal, Chitta Baral. [Hypothesis Generation for Materials Discovery and Design Using Goal-Driven and Constraint-Guided LLM Agents](#). *Findings of Association for Computational Linguistics (NAACL 2025)*
- Neeraj Varshney, Satyam Raj, **Venkatesh Mishra**, Agneet Chatterjee, Ritika Sarkar, Amir Saeidi, Chitta Baral. [Investigating and Addressing Hallucinations of LLMs in Tasks Involving Negation](#). *NAACL@TrustNLP, 2025*

WORK EXPERIENCE

Graduate Teaching Assistant, Arizona State University • Conducted lecture sessions as Instructor-on-Record for students of CSE 485: Capstone I • Supervised and graded 30 capstone teams across multiple assignments and team meetings	August 2025 - Present
Graduate Research Assistant, Cogint Lab, Arizona State University • Developed an LLM-based multi-agent framework to improve tool-use capabilities in customer-service simulations • Spearheaded the research for the systematic study of common reasoning errors occurring in LLM-generated rationales to predict legal judgment scenarios.	June 2024 - Present
Research Aide, W.P Carey School of Business, Arizona State University • Collected, cleaned, and analyzed datasets using Teradata RDBMS and Excel to derive research insights. • Employed NLP models to preprocess datasets through anonymization of personally identifiable information.	July 2023 - Nov 2023
Software Consultant, Oracle Financial Services Software, Bengaluru, India • Engineered the business logic for 30+ requirements of FLEXCUBE using PL/SQL, JavaScript and Java across 10+ Scrum sprints and 3 release cycles. • Streamlined automation scripts in the software deployment cycle, reducing deployment time by 30%.	Sep 2020 - Dec 2021

ACADEMIC PROJECTS

Enhancing LLM Safety Alignment via Augmented Prompt-Response Finetuning • Leveraged prompt-engineering techniques to create a synthetic dataset of 1000+ harmful prompts and corresponding safe responses for injection into an instruction-tuning dataset for LLM alignment. • Augmented Supervised Finetuning of Llama-2-7B LLM with Alpaca dataset and the prompt-response pairs. • Utilized GPT-3.5-turbo and Vicuna-33B as LLM-as-judges to evaluate the finetuned model outputs.
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SKILLS

Programming Languages: Python, C++, Java, PL/SQL, JavaScript Libraries and Tools: PyTorch, HuggingFace, Git, OpenAI, Claude Code, Linux
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